

UNIVERSITY OF CAPE COAST

**ORGANIZATIONAL SOCIAL AND INTERACTIVE
APPLICATION SYSTEM (OSIAS) USING WEB
TECHNOLOGIES IN AN INTRANET SYSTEM**

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ABSTRACT

Majority of faculty staff members, lecturers and students in various departments in Ghanaian Universities spend a lot of time moving from one office to another in order to communicate with each other or send files and documents and other multimedia assets. This project focuses on solving this problem by developing an intranet application that is capable of supporting instant chat, file transmission including downloads and uploads, and other functionalities that are necessary for the success of the system.

Organizational Social and Interactive Application System is a web-based system which uses web technologies that is accessible on an intranet architecture. The application can be accessed using a web browser on a PC, mobile phone and etc. The application is developed using open source web technologies like HTML, CSS, typescript for the front-end and nodeJS, JavaScript and mongoDB for the back-end.

The application developed in this research has functionalities that will enable students, lectures and staff members of various departments in a particular institution to communicate effectively and efficiently.

DEDICATION

I hereby wish to dedicate this project work to my parents, supervisor and lecturers for their love and support.

ACKNOWLEDGEMENT

I am indebted to my supervisor, Mr. Elliot Attipoe of the department of Computer Science and Information Technology, University of Cape Coast for his guidance and direction. His constant encouragement, valuable criticisms, suggestions, corrections where due and patience. May the Good Lord replenish your strength and wisdom.

Special thanks goes to the entire staff of the department of Computer Science and Information Technology.

I also want to thank my family and all my friends who helped me during this study.

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CHAPTER ONE

INTRODUCTION

1.1 Background Study

Intranet Application system is one of the most important application systems for most businesses and cooperations. Intranet application programs are usually outside the world of the internet, which is the biggest collection of interconnected networks. Intranet application are local and only accessible by authorized individuals within a particular organization. Intranet Application offers many unique services that are geared towards a particular problem an organization faces. These problems might be security, time, cost, storage, compatibility and convenience problems. These stated problems give many organizations no other choice than to develop or implement their own systems. Some organizations may decide to allow other legitimate interested parties who are outside the organization like their stakeholder to have limited accesses to the intranet system thereby extending their services to other individuals who might want to know what is going on in the organization.

Today, many institutions and organizations have communication technologies already set in place. These communication technologies include, Wireless Access Point (WAP) also known as Wi-Fi which is a signal that allows the dissemination of data and information. Also, a collection of servers which consist of DHCP server, which is responsible for automatically assigning Internet Protocol (IP) addresses to

computers and devices who may want to connect to the system. Again, a DNS server, which is responsible for resolving address names into IP addresses and lastly, web server, which is also capable of providing and serving web contents like text and pictures to concern individuals. Moreover, some large institutions and organization may have a router, which is responsible for connecting these large organizations and institutions together enabling easy access to resources and enhancing communication.

This study involves individuals who may be indirectly or directly associated with the system. The following are the parties involved in the study; the organization or institution is the key stakeholder to the system, they provide most of the facilities which are paramount to the development of the system, without them, there is no system. The individuals within the organization who interact with the system are also key stakeholders. These particular stakeholders are important to the organization and this study is done to help improve their communication, interactivity and social activities.

OSIAS is an intranet application which provides solution to some communication problems and improves better interactivity and social activities within an organization. There are other existing systems which are similar to OSIAS. These existing systems include Softros LAN Messenger and Win popup LAN Messenger, just to mention a few. These systems all uses the intranet as its main network system. They allow users of the system to communication with each other using

texts, pictures, videos and other multimedia materials. They also provide secure communication for individuals who want to use the system. Again, they offer individuals extra communication functionalities by allowing them to communicate in groups. Lastly, these systems provide basic information about whom you are communicating with by displaying the name, picture and other profile information. The above are all functionalities of the existing systems but there are more room for improvement. It is known that proper communication and good social interaction within an organization yield more and better productivity. My study is to help implement and improve communication and social interaction using intranet technology within an organization and also provide a better solution to improve the existing manual systems.

1.2 Problem Statement

According to my study, the main problem that I encountered was the gap that resulted from the comparison of my proposed system and the existing systems. This gap consists of the issues the existing systems could not solve. These issues include security, time, convince, storage, compatibility and feature enhancement problems. From the study, I realized that not all the existing systems implement security, some existing systems are slow, others are difficult to use and understand, some do not offer options to save conversation and multimedia materials, others do not allow direct installation on other operating systems or platforms, poor design structure

and layout and lastly, most existing systems hardly do upgrades or updates and feature enhancements.

1.3 Proposed Solution

OSIAS is a system to help improve communication and interactivity within an organization using web technologies in an intranet system. The system uses the following web technologies; Two-tier architecture, sockets, network architecture, web programming languages and various protocols. The system also consists of various hardware and connectors like servers, switches, wireless access point, network cables and routers (if only necessary). The proposed system is an improvement to the existing systems, as such, the proposed system consists of all the features of the existing systems with improved functionalities and some extra additional enhancements. These improved features include secure private and group chat, faster transmission of multimedia files, organized file repository, enhanced and validated user identification, better interface design, and lastly, user's chat conversations are saved.

1.4 Aims

The main aim of OSIAS is to offer improved communication and interactivity within an organization in an intranet system. Other aims are to establish secure

inter-connectivity between individuals, and also to provide easy to use, time effective communication and interactive system.

1.5 Objectives

To accomplish and successfully meet the aims of the project, these following objectives will help track the progress of the project;

- Development of a sub-system which will store the individual's media files.
- Also, the development of a sub-system which will be capable of allowing users and interested parties to download and upload media files.
- Again, the development of a sub-system which will monitor the individual user accounts e.g. signing and logging into an accounts.
- Lastly, the development of a sub-system which will store and manage the various functions of the system like chat and post.

1.6 Significance of Study

In this research project, the focus was on the important functions and features of OSIAS such as secure private and group chat, faster transmission of multimedia files, organized file repository, enhanced and validated user identification and lastly better interface design. These functions and features within the OSIAS provides so many avenues for the system to be used in many different organizations and also to be integrated into other application areas such as in schools, universities, public

libraries, business offices and scientific organizations, just to mention a few. The proposed system can be a future replacement for many internet and intranet chat applications and other similar systems, this is because it costs less to implement, it is secure, it is fast, it is compatible and easily integrated with existing legacy system, it is better in performance in many aspects as compared to other systems and lastly it is convenient for an organization to integrate OSIAS into their existing systems.

1.7 Organization of Study

This proposed system is directly geared towards academic institutions. OSIAS can be used in other organizations but it will need some minor refactoring like the contents (texts) while still its functionalities and features being intact. The communication and sociability of OSIAS is to provide a greater means for academic officers like the lecturers and staff members to interact and communicate with each other whiles the interactivity aspect of OSIAS is to provide avenue for better lecturer-stuff relationship and interaction.

1.8 Structure of The Study

This project is divided up into six chapters and each chapter describes a separate aspect of the work done in the course of the project.

Chapter one is the introduction to the project topic. This deals with what the project topic is about, the problem statement, the proposed solution, the aim and objectives, the significance of the study, the organization of study and lastly, the structure of the study.

Chapter two is the literature review of the project topic. This includes information pertaining to the current knowledge on the project like the importance of having a chat application in an organization, the various architecture and their various functions and advantages, the technical aspect of the system like the socket protocols and others and lastly, how the system works in comparison to already existing ones.

Chapter three talks about system requirements and specification. This chapter deals with the various requirements that are necessary for the system to run optimally. This consist of the software and hardware requirements and also the expected functionalities specified by the user or interested entities. This is called the user requirement.

Chapter four emphasis on the software overview which consist of system architectures like database and technical architectures, the user documentation and manuals, the various necessary models and diagrams like the use case models and lastly the ideal implementation methodology used for the project which is the Rational Unified Process.

Chapter five elaborates on the testing of the system. This involves the steps taken to test the system for any unresolved errors and bugs. The testing approach used were alpha and beta testing and quality assurance testing.

The last chapter, which is chapter six deals with summary, conclusion and lastly the recommendation.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Electronic communication is getting more and more popular among public and private sector organizations. Apart from using telephones or automobiles and sending mails, people can now communicate and interact with each other through the chat technology. The chat is a kind of electronic technology that supports human-to-human communication and interaction. The following illustrates the importance of having a chat system in an organization or institution. It is faster than other means of communication like the postal and Email system. It is also convenient and easy to interact with. It costs less to implement and lastly, other manual mean of communicating like the paper system harms the environment.

2.2 Three-Tier Architecture

The OSIAS uses three-tier client server architecture. A three-tier architecture is a client-server software architecture pattern in which the user interface (presentation), functional process logic ("business rules"), computer data storage and data access are developed and maintained as independent modules, most often on separate platforms. The application handling is completed separately for

database queries and updates and for business logic processing and user interface presentation.

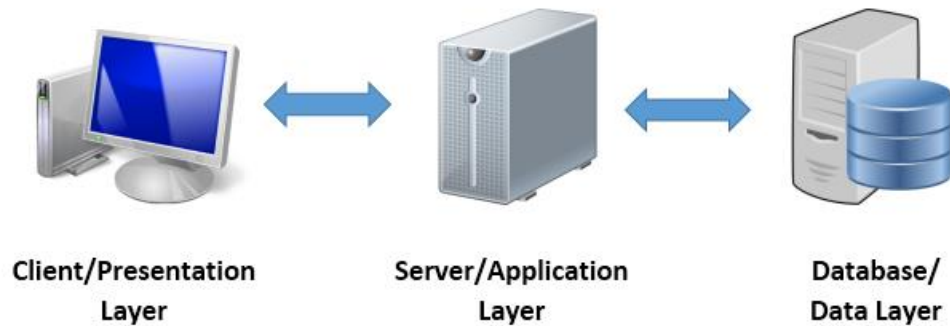


Figure 2.1 Three Tier Architecture (SunilwanJayathissa, 2015)

Generally, an intermediary server called the web server bind the back-end of an application to the front-end, though all tiers can be present on the same hardware. The three-tier design generally encompasses client demanding services from server the web server which intern query the database server for services (Chandra & Kumar, 2009).

2.3 Socket

In order to implement OSIAS, it needs socket which is an object that represents a low-level access point to the IP stack (Reid, 2004). A socket is a simulated medium that allows sending and receiving of data in an application. The socket enables an application to connect to a network and communicates with other applications connected to that same network (Donahoo & Calvert, 2009). A socket basically constitutes the combination of an IP address and a port number.

(Socket = IP address + Port Number).

There are many types of sockets that fit into dissimilar protocols and dissimilar stacks of protocols in a suite. However, this paper is focused on the TCP/IP protocol because the TCP/IP is the standard and mostly used protocols in instant messengers. The primary or fundamental sockets in the TC/IP are the stream socket and the data-gram socket (Donahoo & Calvert, 2009).

Based on the types of TCP/IP sockets and how they work, in this project, I employed the stream socket because it uses the TCP protocol which allows limitless messages to be sent over a network unlike the data-gram socket that is limited to 65,000 bytes. Besides, it is very secure and more reliable as compared to the data-gram socket. Also, it ensures reliable delivery of messages and files.

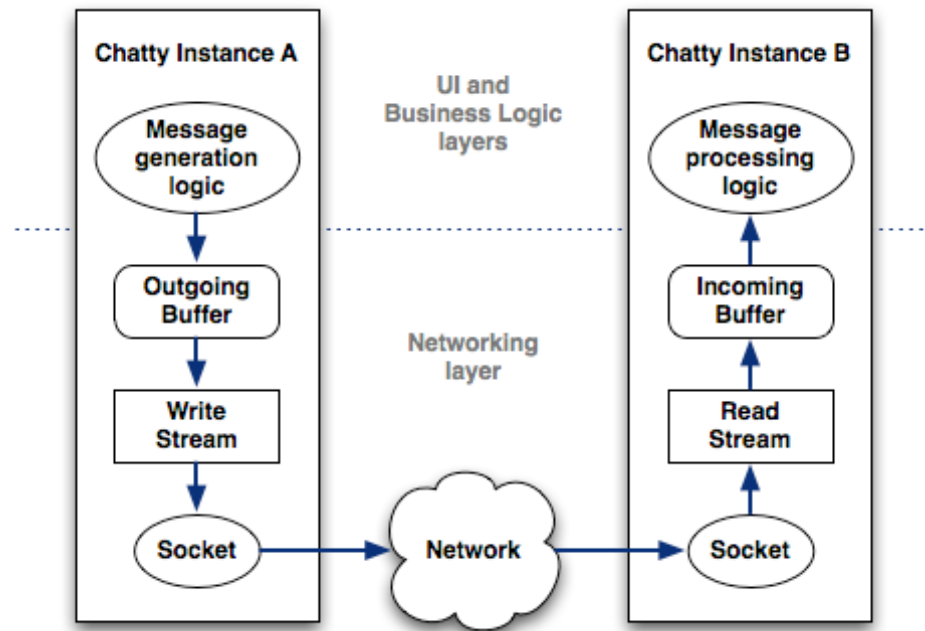


Figure 2.2 Socket (Bakhyryev, 2018)

2.4 How Chat Communication Works

Two sockets are created at the client side and the server side. The client connects to the server through its IP address and port number. They must share the same port number for them to communicate (Reid, 2004). The client and the server both communicate through a stream of bytes written to the socket. The client and the server must agree on a protocol (TCP, UDP or RAW) and agree on the language of the information transferred back and forth through the socket (Umar & Justin, 2003). Basically, the message sent by one staff into a socket and passes it to another staff on the receiving side. If it is group chatting, a central socket will be used to collect the message and then it will be broadcast to all staff that are active.

2.5 Network

Peer-to-peer network is a type of network in which interconnected nodes share resources amongst each other without the use of a centralized administrative system. A client/server network is a type of network where individual clients request services and resources from centralized servers.

The below table provides a vivid comparison between a client/server network model and peer-to-peer network model.

	Peer to Peer Network	Client/Server network
Hardware Cost	It needs no high-end server as the resources are distributed over all clients which reduce cost.	A dedicated computer server (hardware) that distributes resources is needed.
Easy Setup	It is easy to setup mainly if the computers are less than fifty (50).	It is difficult to setup.
Network Operating system	There is no required network operating.	Network operating system is required.
Failure	It can accommodate failure i.e. if one or more Computers (clients) fail the others can still be up.	It cannot accommodate failure if the server fails.
Security	It has security deficiency as clients' administration is not guaranteed.	Very secure because server administration is guaranteed.
Performance	It performs less	Performs very good
Backup	It has decentralized backup that is difficult to access.	It has a centralized data backup with ease of access.

Table 2.1 Comparison between peer-to-peer and client-server architecture

Client/server network was used to implement the OSIAS because of the central administration, the scalability, this mean that, any component can be upgraded when required, and lastly, the flexibility, this deals with how new technology can be integrated into the network if the client should increase.

2.6 Existing Systems

There are many existing local area network chat applications which offers multiple functionalities. The following are the only two already existing local area network chat applications that I researched on prior to the execution of my project. The first is Winpopup Local Area Network Messenger and lastly, Softros Local Area Network Messenger.

Winpopup messenger cooperation strongly believes in a good interrelationship between the employees with an organization and as such continuously tries to improve and maintain the level of communication by coming up with new solutions and better functionalities.

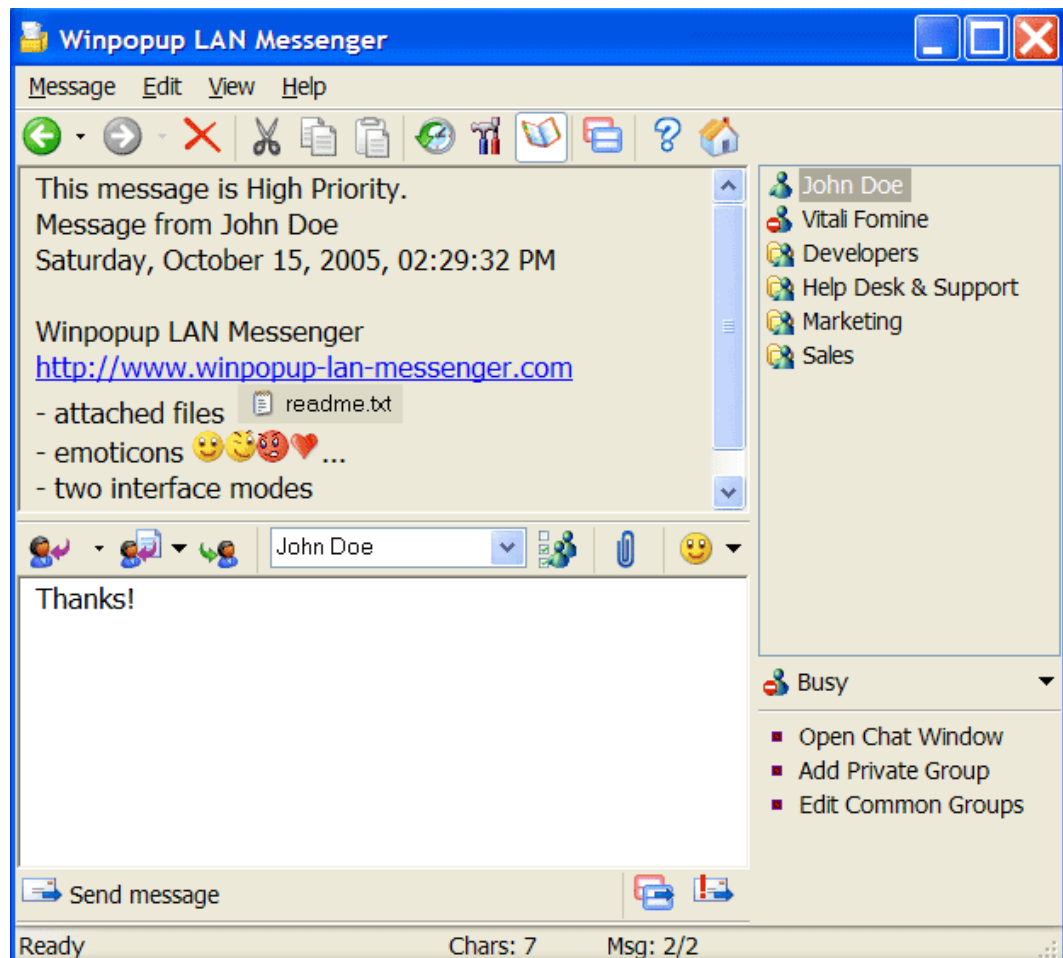


Figure 2.3 Winpopup Messenger

According to Winpopup messenger website, they stated that, when one operates a small business, then you have no problem communicating with all your employees

all at once. When your company grows, becomes more and more difficult to communicate. You either run across the building to ask a few questions to several employees, or waste precious time setting up a common meeting. You can ask the same questions by email, but in the modern days of accelerated pace of work in a typical office environment, email is plain too slow. If you have a Local Area Network, then you possess a resource that is capable to deliver instant messages to your employees in zero time and it is just the time to consider a corporate instant messaging solution.

Again, they emphasize on the fact that their system needs no internet connection, saying that, the availability of the internet connection in a local corporation used in communication creates security threats in the network and also the unavailability of the internet can cause the system to fail.

All communication remains within the local network area, never leaving the secured intranet environment. Moreover, they elaborate on some of the functionalities of their system, saying that, in case there is an urgent group meeting, there is no faster way to call concerned users or employees than to use the group announce feature of the Winpopup Local Area Network Messenger, which only requires selecting a group of employees and typing the message you want to send, and it will be instantly delivered to the entire group.

Furthermore, they talk about how their system can be use to do remote group meetings and discussions on the Local Area Network, saying that, suppose there is an issue or a quick question, but no one is available physically to solve, one only needs to ask the question to the group and since Winpopup Local Area Messenger supports group broadcast and group chat, the entire group can read and discuss the question and offer suggestion and solution in real time. Lastly, they talk about how easy it is to learn. They stated that, Winpopup Local Area Network Messenger supports extended and simple views, the latter bearing resemblance to well-known messengers such as ICQ or MSN Messenger which most computer users are very comfortable with these stated products, and as a result will not spend much time to master the system.

According to Softros Local Area Network Messenger (2018), they define their system as an easy-to-use Local Area Network messaging application which is safe, secure and effective for intra-office communication. They stated that, it does not require a server to run and it is very easy to install which comes with a variety of handy features such as PC-to-PC messaging, group Local Area Network chat rooms, broadcast messaging which quickly notify selected individuals or groups about an even and also drag-and drop file transfer to exchange file and folders between staff members.

They also stated that, all messages and files exchanged by the users are securely encrypted and never go outside the local company network, which guarantee that

no unauthorized person will have access to sensitive data. Since all data is encrypted with AES-256 encryption algorithm by the software which only works strictly within the company network, it guarantees for the security, safety and privacy of the system and data. This as a result, allows employees to safely exchange messages and files, while discouraging and preventing them from chatting with people outside the organization. Also, according to Softros, this helps to save internet bandwidth and minimizes the number of opened firewall ports that could result in outside hacker attacks, virus and worms.

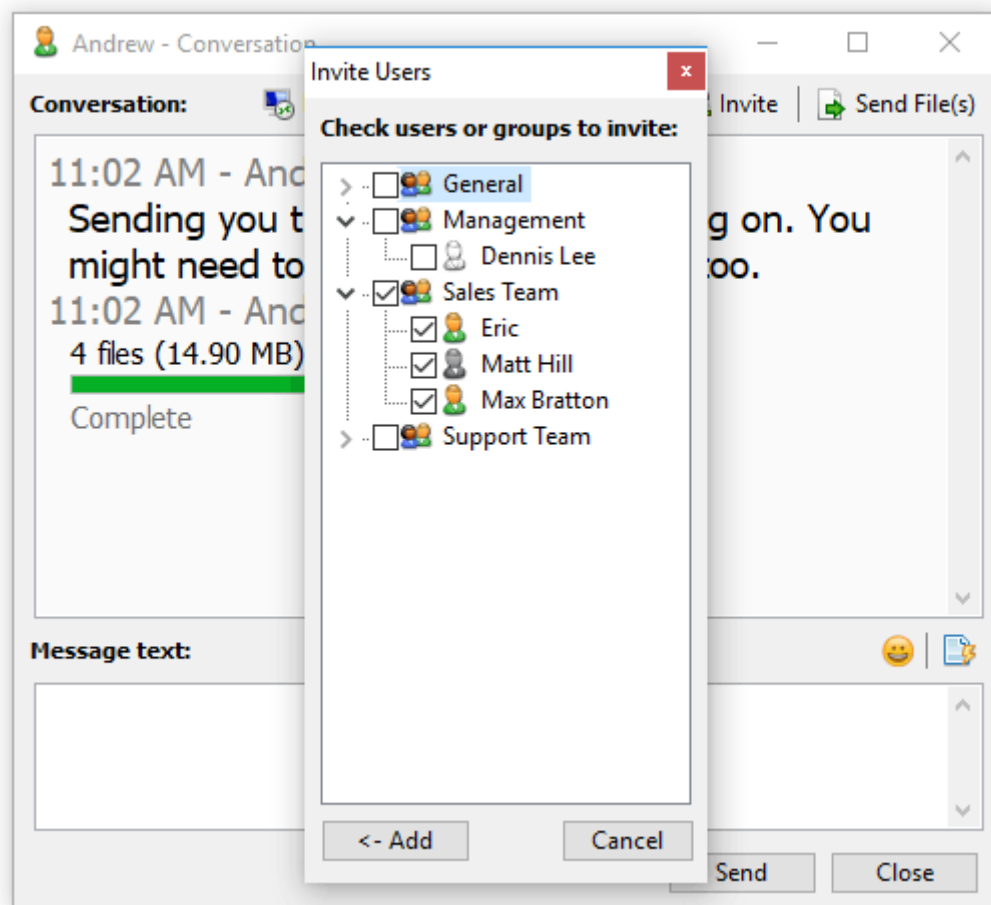


Figure 2.4 Softros Local Area Network Messenger

They also elaborated on the ease of installation and use, according to them, Softros has a very intuitive interface and requires no additional training, all you need to do is install and run it on each network computer and users themselves can easily configure the program without assistance from the Information Technology staff. Other features they provide include server-less architecture, which simply means that, there is no need to set up and maintain a dedicated messaging server, in a technical sense it is based on a peer-to-peer network architecture. Also, the system caters for individual and group conversation, this means that, exchange of messages can occur only between yourself other single user or with other multiple users at the same time. Moreover, according to Softros, they stated that, their system allows for group messaging, which notify individuals users or groups about an event by sending out a broadcast message. Also, it allows users to send messages to other users even when they are offline. Offline messages are received when the user comes online.

Again, file transfers are allowed by the system which involves easy exchange of documents with colleagues. This can be achieved by just dragging and dropping a file or folder on the recipient's icon in the user list or on the conversation. Other functionalities which I will not explain further include; Remote Desktop Sharing, Message History, Terminal Services Support and lastly Restricting user access to certain resources and functionalities.

CHAPTER THREE

SYSTEM REQUIREMENT AND SPECIFICATION

3.1 Introduction

All computer software needs certain hardware components and other software resources to be present on the computer. These prerequisites are known as system requirements and are often used as a guideline as opposed to an absolute rule. A second meaning of the term system requirements, is a generalization of this first definition, giving the requirements to be met in the design of a system or sub-system. All systems requirements define two sets of requirements, these are minimum and recommended.

Recommended System Requirements are set of requirements that are different from those that are needed to run a software. They are almost always of a significantly higher level than the minimum requirements, and represent the ideal situation in which to run the software. Generally speaking, this is a better guideline than minimum system requirements in order to have a fully usable and enjoyable experience with the software.

Minimum System Requirements are the set of requirements that are highly needed to run the program. They are the least of all requirements. These requirement does not necessary mean the software will run optimally. It generally provides a

benchmark for the least of all requirements that are necessary for the specific program to run but not in its prime conditions. These requirements sometimes do not provide the enough enjoyment for the users running the program.

Minimum Hardware Requirements

Computer/Processor	Intel® Core 2 Duo (1.6 GHz) or equivalent AMD Athlon 64 X2
Memory (RAM)	3 GB, more memory improves performance
Hard disk	20 GB of free hard disk space
Optical Drive	Internal/external DVD x8 drive
Display	XGA (1024 x 768 pixels) or higher resolution monitor with 16 bit color settings
Input device	Keyboard and mouse or compatible pointing device
Internet connection	Network using 10 Mbps or higher speed network adapter and/or WLAN and/or 33.6 Kbps or higher speed modem (not supported for Win 7/8 clients).
Miscellaneous	USB interface required. A battery-pack that may provide 500 mA via USB.

Table 3.1 Minimum Hardware Requirement

3.2 User Requirements

The user requirement is most considered as a document which is used in software engineering, this document specifies what the user expects the software to be able to do. User requirement usually starts with information gathering, which is then

documented. These information spell out exactly what the software must do. A user cannot demand features not in the user requirement document, whilst the developer cannot claim the product is ready if it does not meet an item of the User requirement documentation.

Why is User requirement documentation important? This is mostly important in three main situations. The first is that, it serves as a guide to planning cost, milestones, timetables, testing, etc. Also, it allows developers to show it to various stakeholders to make sure all necessary features are described. And lastly, it helps to determine what is technically and economically feasible.

User Requirement Documentation usually include priority ranking for each requirement. This priority ranking describes the degree of urgency of the various requirements. A typical user requirement priority ranking might include the following; Mandatory requirement, this include features that must be built into the final system. Another is Desirable requirement; this include features that should be built into the final system unless the cost is too high. Also, Optional requirement, this consist of the features that can be built into the final system only if the developer wants to. Lastly, Possible future enhancement, this include features the developer may want to build into the final system, however, the developer just wants the feature or features in and of itself.

3.3 Hardware Requirements

The most common set of requirements defined by a software application is the physical computer resources, also known as the hardware. A hardware requirements list is often accompanied by a hardware compatibility list (HCL), especially in case of operating systems. An HCL list tested, compatible and sometimes incompatible hardware devices for a particular operating system or application.

All computer operating systems are designed for a particular computer architecture. Most software applications are limited to a particular operating systems running on a particular architecture. Although architecture-independent operating systems and applications exist, most need to be recompiled to run on a new architecture.

OSIAS is an architecture-independent application system, this is because it only runs in the browser and does not make any direct system request to resources. And as such little or no system complication may occur when the application is loaded and ran.

The power of the central processing unit (CPU) is a fundamental system requirement for any software. The above minimum requirement specified in the system requirement section only require a CPU with a processing speed of 1.6

gigahertz or higher. This specification allows for an optimal usability of the application.

All software, when run, resides in the random access memory (RAM) of a computer. Memory requirements are defined after considering demands of an application, operating system, supporting software and files and other running processes.

Hard-disk requirements vary, depending of the size of the software installation, temporary files created and maintained while installing or running the software and possible use of swap space (if RAM is insufficient).

Software requiring a better than average computer graphics display, like graphics editors and high-end display adapters in the system requirements. Any computer system with an embedded graphics card or its equivalence is enough for OSIAS to operate optimally.

Some software applications need to make extensive and/or special use of some peripherals, demanding the higher performance or functionality of such peripherals. Such peripherals include CD-ROM drives, keyboards, pointing devices, network devices, etc. No high end intensive peripherals are required in OSIAS, only the conventional peripherals like monitor, keyboard and mouse and other networking equipment.

3.4 Software Requirements

This deal with defining software resources requirements and prerequisites that need to be installed on a computer to provide optimal functioning of an application. These requirements or prerequisites are generally not included in the software installation package and need to be installed separately before the software is installed. These software include nodeJs and MongoDB application for the respective operating systems.

A computing platform describes some sort of framework, either in hardware or software, which allows software to run. Typical platforms include a computer's architecture, operation system, or programming languages and their run-time libraries. Software may not be compatible with different versions of the same line of operating systems, although some measure of backward compatibility is often maintained. OSIAS is platform independent

Most web applications used in public and private sector organizations which depends heavily on web technologies usually make use of the default browser installed on system. e.g. Microsoft provides privilege system access through the use of their ActiveX module. OSIAS do not depend on any module and it works on all browsers, but i will prefer Chrome browser as the default browser.

CHAPTER FOUR

SOFTWARE OVERVIEW AND IMPLEMENTATION METHODOLOGY

4.1 Introduction

This chapter seeks to explain how to operate or how to use the software and the other various requirements and prospectus that provide an inside explanation to the components that make up the system. These requirements and explanations may include the architectural design, the technical design, and lastly the end-user manuals. This chapter also briefly highlights of the implementation methodology that was used in developing the system.

4.2 Architecture Design

This is also known as the software architecture description. This highlights the general requirements of the various routines that exist in the software. This provides a short detail description and explanation of the various component of the system. These detail descriptions consist of the technical design architecture, user interface and the programming codes used in developing the system.

Technical Design Architecture consists of the code documentation associated with the source code which may include the README files, API documentations and sometimes the various routine comments within the source code. These

documentations may be used by the developers, testers, and also the end-users using the application.

User documentation and manual simply describe how a program is used. It describes each feature of the program and assists the user in realizing these features. There are three broad ways in which user documentations can be organized, these include tutorials, thematic and list or reference. I will concentrate more on the tutorial approach to user documentation. Tutorial approach provides visual and/or audible guide through which users can follow in a sequential manner to accomplish a particular task. This approach is considered the most useful for new users as it is very easy to comprehend. Composing user documentation consist of five steps, these include User analysis, Planning, Draft review, Usability testing and lastly Editing. These are all preformed in a particular order.

4.3 Use Case and Use Case Scenario

A user case represents the actions that are required to enable or abandon a goal. It consists of multiple paths that can be taken by any user at any one time (Felder, 2007). This provides a clearer outlook on the scope of the project. This also describes the process of a system by using actors and the typical workflow. A use case scenario is a single path through the use case.

The various actors for this system are as follows; Lecturers, Administrators and lastly, the Stuff Members and the typical workflow include post text, post media file, filter post, chat with colleagues, share media files with colleagues, create and delete accounts, and lastly restrict and allow permissions. A typical use case scenario diagram is a follow.

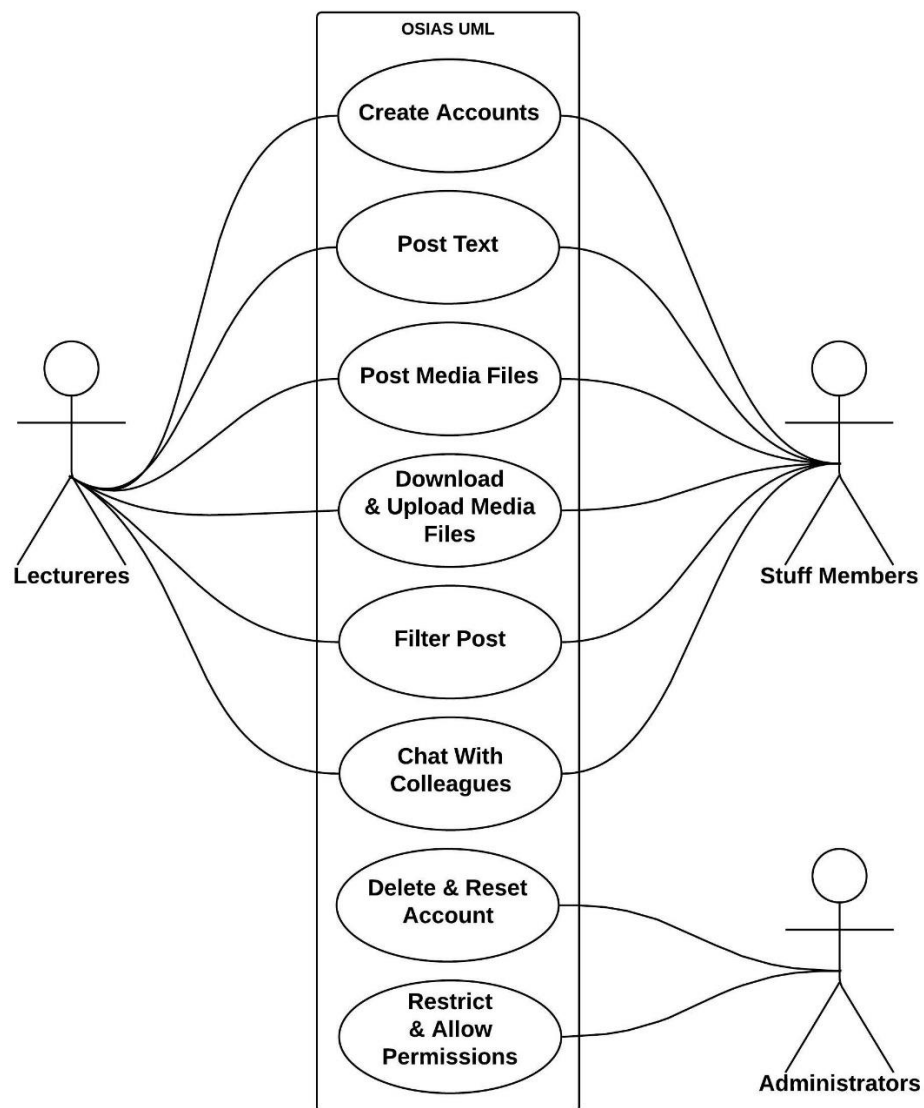


Figure 4.1 Use case diagram

4.4 Site Map

According to techopedia (2018), A site map is a model of a website's content designed to help both users and search engines navigate the site. A site map can be a hierarchical list of pages (with links) organized by topic, an organization chart, or an XML document that provides instructions to search engine crawl bots. The diagram below is a sitemap for Organizational Social and Interactive Application System. Users will have access to different pages depending on user access level.

figure sitemap

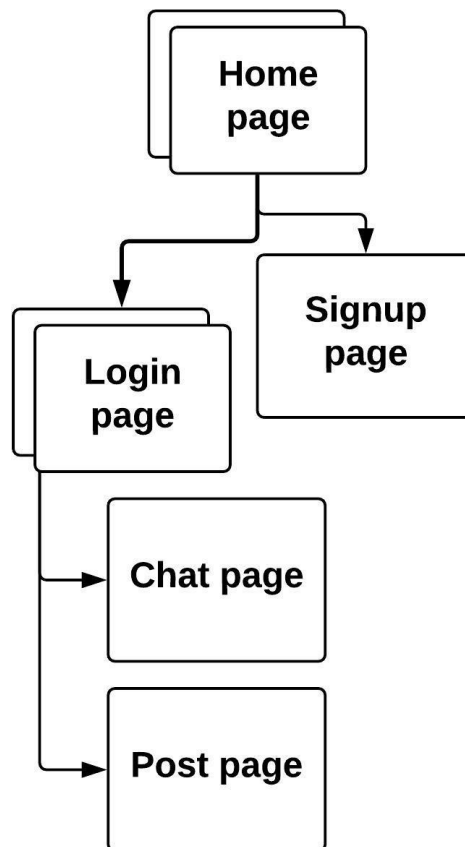


Figure 4.2 Site map

4.5 Database Architecture

This consist of the conceptual, logical, and physical design elements. This provides formal information to people who may interact with the database. The purpose of preparing it is to create a common source of information and instruction which helps database designers, developers, administrators and application designers to interact with the database system. It also provides information on how to create, update, insert or delete entities or collections in the database as changes are always bound to happen.

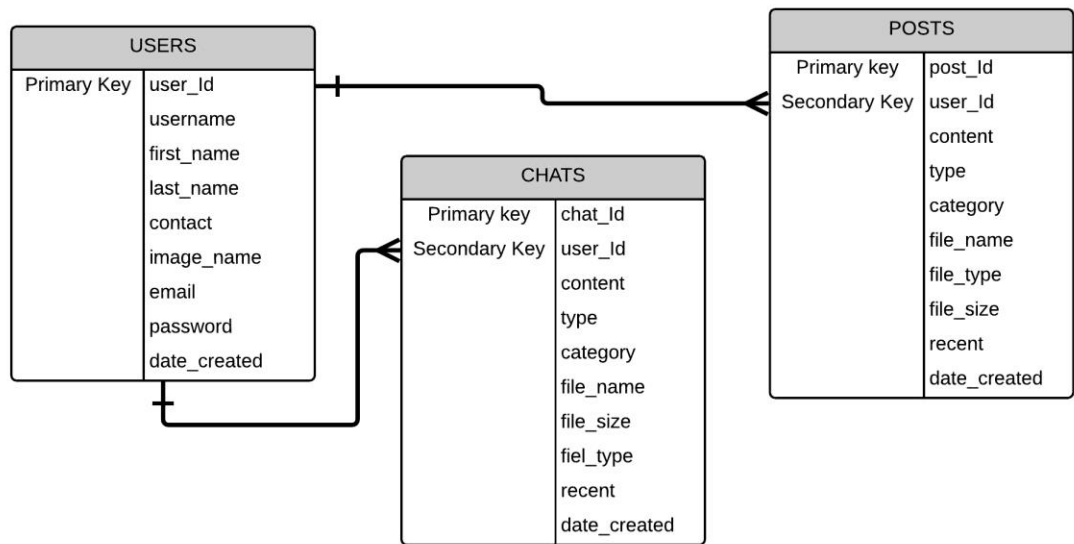


Figure 4.3 Database schema

OSIAS's database architecture uses a no SQL paradigm. MongoDB is a no SQL paradigm database architecture system. It is procedure in nature, it can also be object-oriented. The main programming language for MongoDB framework is

JavaScript. The below Figure highlights on the various relationships and cardinalities between the entities or collection with the database.

4.6 Flowchart

A flowchart is a type of diagram that represents an algorithm, workflow or a process. It shows the steps as boxes of various kinds, and their order by connecting the boxes with arrows. This is used in analyzing, designing, documenting or managing a process or program in various fields. The diagram below represents detail flowchart of the system. Illustrated in Figure 4.

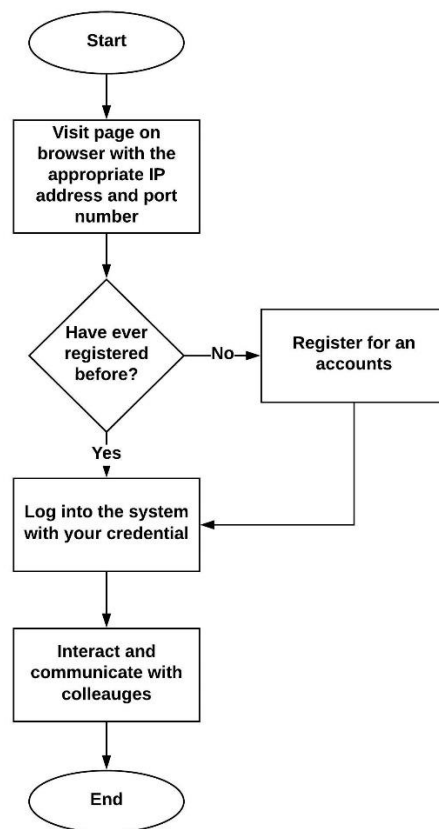


Figure 4.4 Flowchart

4.7 Implementation Methodology

It is the carrying out, execution, or practice of a plan, or method, or any design, idea, model, specification, standard or policy for doing something. This involves the software development methodology used in implementing the system. This also provides a framework that is used to structure, plan and control the process of developing a system.

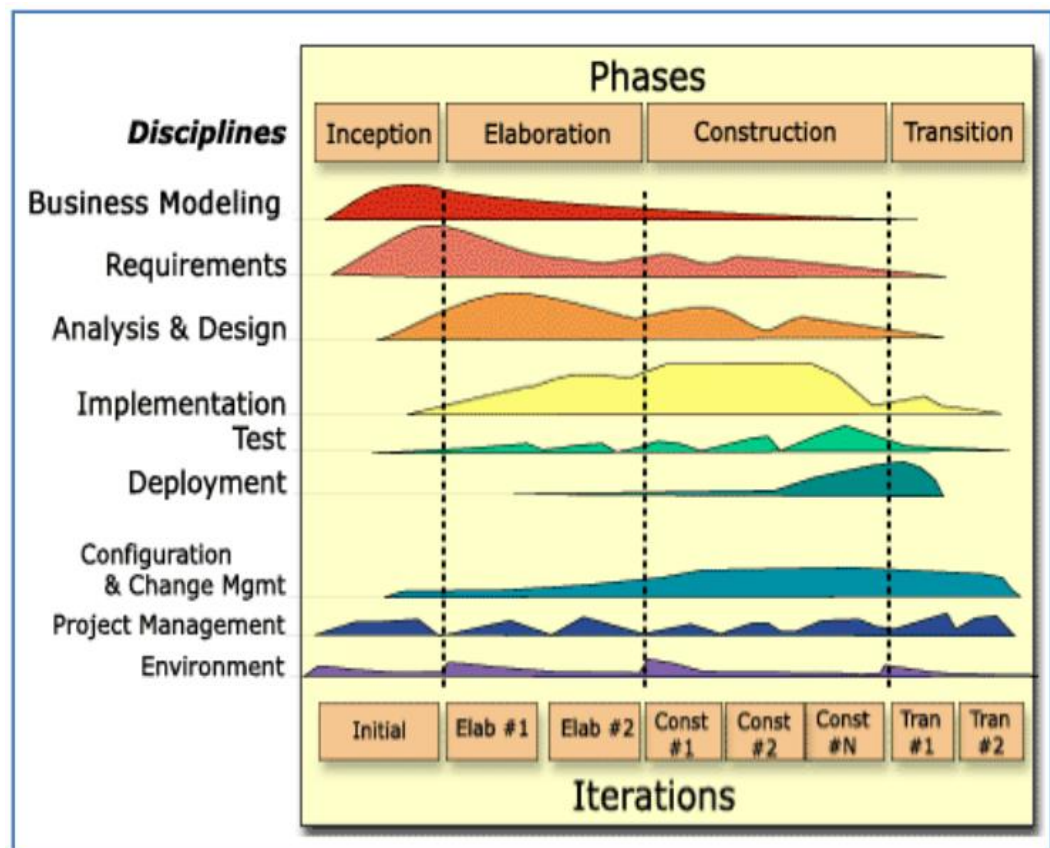


Figure 4.5: Rational Unified Process (Zsolt, 2018)

The implementation methodology used in OSIAS is Rational Unified Process (RUP). This process attempts to capture many of modern software development's

best practices in a form suitable for a wide range of projects and organizations. This process is a type of iterative approach to software development.

Rational Unified Process uses the following key steps or process which are Inception, Elaboration, Construction and lastly Transitioning. These process or steps are also called phases in Rational Unified Process. Below are the detail description of the various phases.

The Inception phase is the fundamental process of understanding why an information system should be built. It also enables the developer or designer to determine how the project will transgress when building the system. During project initiation, the business value of the project to the organization is identified. At this phase, scope of the project is also defined and its business case.

In the Elaboration phase, analysis of project requirement is done in greater detail. This phase enables the developer or designer to define what the functional and non-functional requirements of the system will be. Also, this phase helps the designer or developer to define the aim, feasibility and plan of the project. At the end of this phase the developer reviews the requirements to see if these are complete. If it does not, then the necessary corrections are made, else, the developer proceed to the next phase.

The construction phase consist of the software design. At this phase, source code is also produced. The developer mostly focus on designing the high-level design, the database, the interface and the detailed logical design which make up the system and observe the rules governing design principles i.e. usability principles. The high-level design describes the structure of the data and the algorithm required to implement a module of the system. The database design gives a detailed description of the data schema required to support the high-level design, while the interface design will describe how the human-computer interaction of the system will be. At the end, the developer reviews the design. If flaws are observed, then the necessary corrections are made. At the completion of the design phase, the implementation phase begins. The developer transforms the high-level design into codes and integrate the database design into the codes. After completing the phase, the developer then reviews it and proceed to the next phase. Also in this phase, the developer separately test each of the modules to find and fix any bug using outlined white box and black box test criteria. Thereafter, the developer integrates the tested modules and components together. After this, the developer test the entire system again to find and fix bugs that may result.

Lastly, final system is tested and put into operation at the transitioning phase. After the system has been installed and in use, environment definitely changes, which will need the system to change in accordance to the environment change. So in this phase, when such changes e.g. new technologies, new methods, and etc occur, the developer or other designers that were not involved in the development of the

system have to upgrade the system to meet the changes of the environment using the proper documentations of the system.

Why Rational Unified Process? This process provides efficient and effective means to the development of the software. The following are the reasons for using Rational Unified Process in OSIAS implementation. The first is that it takes into account changing requirements of the system. Another is that, integration of the system is not a big problem such that elements are integrated progressively. Also, it allows for easy identification of common parts as they are partially designed and implemented, this helps to facilitates reuse of codes.

CHAPTER FIVE

TESTING AND DEPLOYMENT

5.1 Introduction

This is an investigation conducted to provide stakeholders with information about the quality of the software product or service. This can also provide an objective, independent view of the software to allow the business to appreciate and understand the risks of software implementation. Test techniques include the process of executing a program or application with the intent of finding software bugs, errors or other defects and verifying that the software product is fit for use.

Software testing investigates the extent to which the components or system under test meets the requirements. It also examines if the system responds correctly to all kinds of inputs. Also to ensure that the system performs its functions within an acceptable time and lastly to know if the system achieves the general result its stakeholder's desires.

5.2 Order of Testing

There are many testing approaches used in analyzing and evaluating the credibility of a system. These testing approaches are outlined testing, unit testing and many more. The following are the order of the testing procedure; outlined testing, unit

testing, bug reports, and lastly, a test summary. This project strictly focuses of the above state testing approach.

Outlined testing approach follows a procedure which ensure that all the functionalities and processes in the system works seamlessly while unit testing deals with the testing of each module or component of the system to ensure that the various components function independent of the rest of the system. Bug reports created during the test cases cover all the faults that resulted in the failure of each test case. A test summary report generated from the testing results helps to mitigate all the defects of the system that surfaced during the vigorous testing procedure.

5.3 Test Plan Objectives

This test plan supports the following objective which ensures and verify that OSIAS works as expected and meets the requirements outlined in the user and software requirement specification.

5.4 Testing Process and Approach

The testing process follows the objectives of the testing plan. The following testing approach were used which includes the functionality test, usability test, interface test, database test, compatibility test and lastly the security test.

The functionality testing tests the specifications and capabilities of the application. The testing activities carried out include the following; testing all the links and forms of the site to ensure that they work correctly and that there are no broken links or faulty forms.

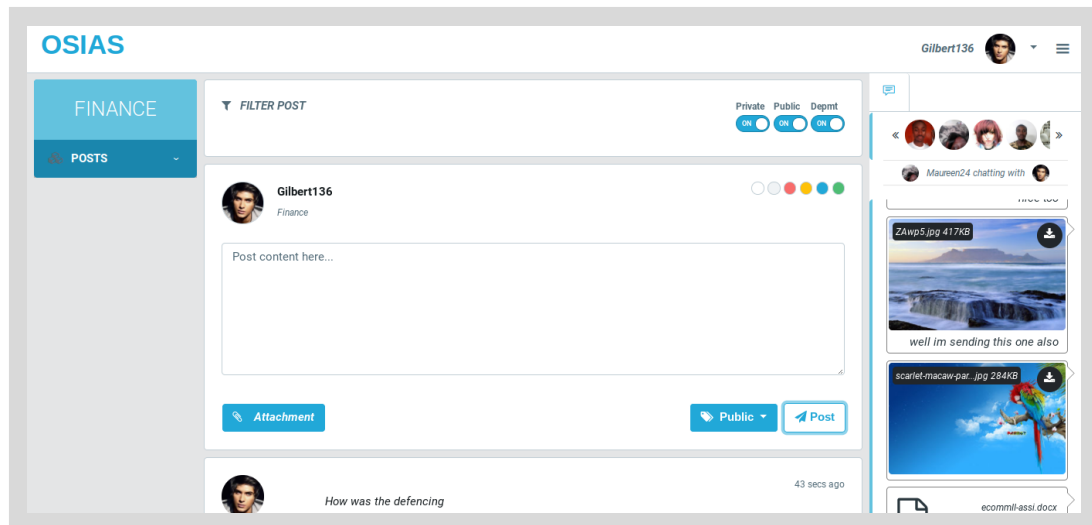


Figure 5.1: Post and chat page

Register

Create your account

	Username
	Firstname
	Lastname
	Email
	Contact
	Computer Sciences & Information Technology
	Please Click Here To Upload Your Profile Picture.....
	Password
	Repeat password

Create Account

Log In

OSIAS © 2018 Copyrights All Rights Reserve

Figure 5.2 Registration or signup page

Login

Sign In to your account

Login

OSIAS © 2018 Copyrights All Rights Reserved

Sign up

Welcome to OSIAS, Sign Up to be a member of this organisation. You can only register once. Please obey the law. If there is any problem, see one of the administrators.

Thank you

Register Now!

Figure 5.3 Login page

The usability testing is conducted to ensure that the system is able to serve its purpose very well, taking into consideration how users of the system may also react to the system. A system may solve a problem but it should not introduce more extensive tasks to the user. The areas covered under usability testing include; user interface content, the system navigation, the legibility and grammatical errors and lastly, the images used throughout the system and the alternative attribute in the image tag of HTML is present, which will be displayed in case there is a slow internet connection or when image loading is being disabled by the user.

The interface testing deals with three major components of the system. This includes the application, the web server and lastly, the database server. The application server interface testing checks if all requests are sent correctly to the web server and the output or response from the web server is displayed in the correct, readable

and legible format by the user. The web server interface testing checks to ensure that all application requests are handled properly without any service denial. This test is done to make sure that all the authorized ports are opened and the application is running. In the case of OSIAS port 4200 and 3000 should always be open in order for the application and the web server to work. The database server interface testing seek to ensure that there are no errors that many result in executing queries which are requested by the application and the web server. This also checks to ensure that that data integrity is maintained while creating, updating or deleting entities, records and fields in the database. Moreover, this test sought to reveal any possibility of data integrity violation data duplicates and others.

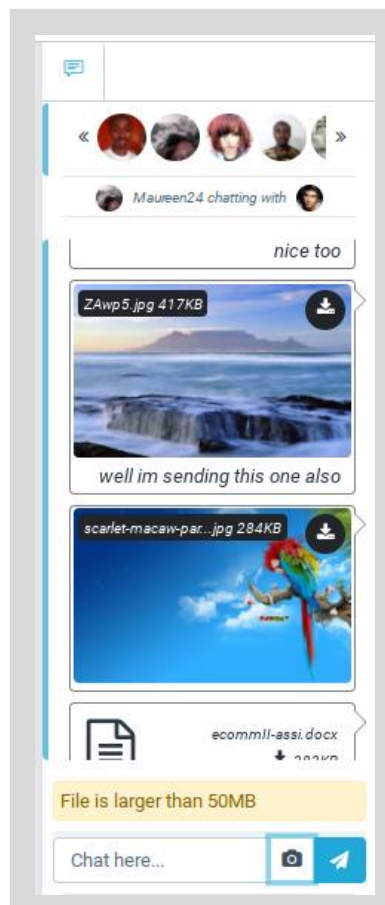


Figure 5.4 File size too big for the chat page

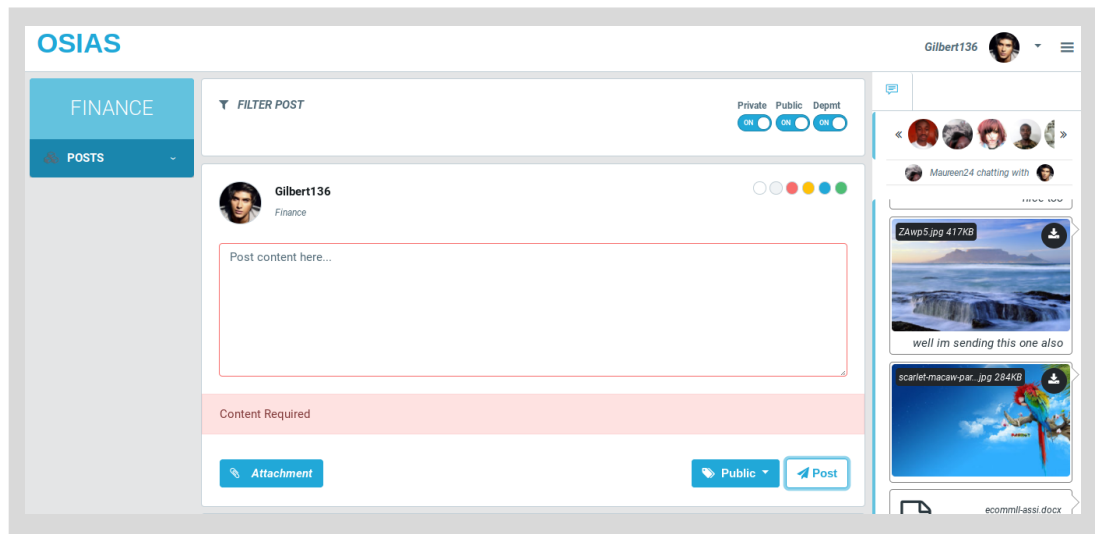


Figure 5.5 Content required

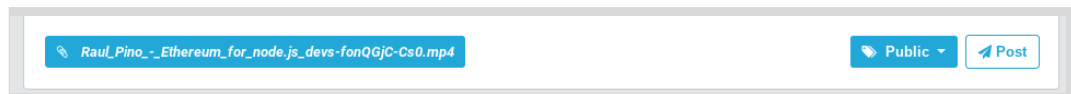


Figure 5.6 File upload

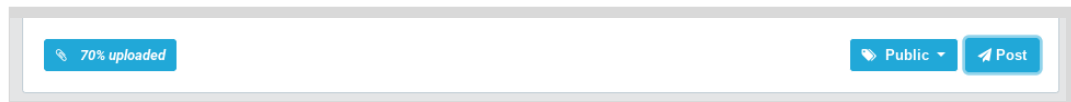


Figure 5.7 File upload progress

Register
Create your account

You are registered and can now login

Create Account

Log In

OSIAS © 2018 Copyrights All Rights Reserve

Figure 5.8 Signup success.

The compatibility testing ensures that the application displays correctly across different devices. This may include; browser compatibility and screen display or responsive display. The browser compatibility deals with the support of the various popular browsers like Chrome, Firefox, Safari, Opera and others. OSIAS is

supported in all the various web browser listed above. The responsive display talks about how dynamic the display interface behaves on different screen sizes and devices.

Login

Sign In to your account

 Username

 Password

Login Register Now!

OSIAS © 2018 Copyrights All Rights Reserved

Figure 5.9 Responsive behavior of the login page

OSIAS



FILTER POST

Private

Public

Depmt

ON

ON

ON



Gilbert136

Finance

Post content here...



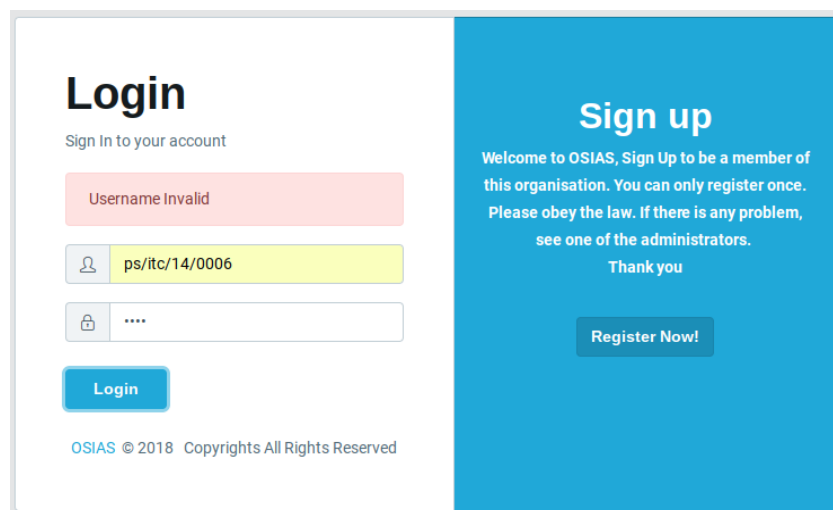
 Public

 Post

» Previous

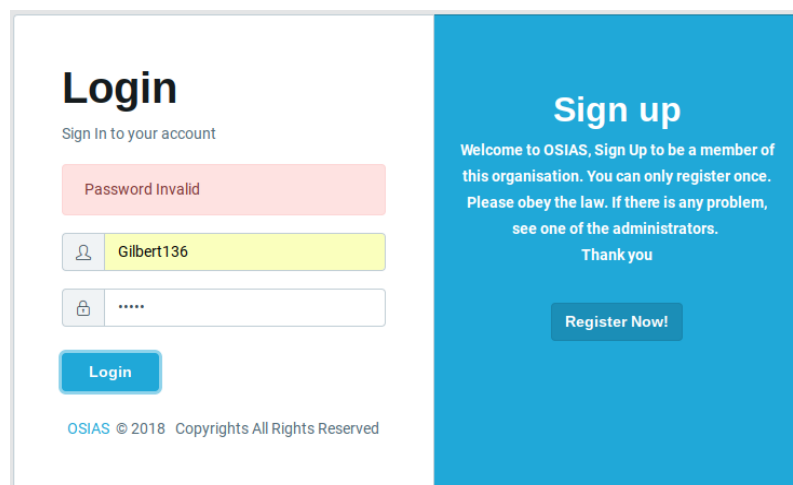
Figure 5.10 Responsive behavior of the post page

The security testing seeks to ensure that the system is solely free from any vulnerabilities and security problems. This test seeks to know how secure the login and sign up component of the system is. Also, this testing is done to ensure that no unauthorized user can gain permission to any page which they do not have access to, e.g. a hacker may gain access to other user's page by just typing in the URL address of that user. Lastly, this testing is done to ensure all users have access to their respective resources.



The screenshot shows a web application with a white login section on the left and a blue sign-up section on the right. The login section has a heading 'Login' and a subtext 'Sign In to your account'. Below this, a red error message 'Username Invalid' is displayed. The username field contains 'ps/itc/14/0006' and the password field is masked with '****'. A blue 'Login' button is at the bottom of the form. The sign-up section has a heading 'Sign up' and a subtext 'Welcome to OSIAS, Sign Up to be a member of this organisation. You can only register once. Please obey the law. If there is any problem, see one of the administrators. Thank you'. A blue 'Register Now!' button is at the bottom of the sign-up section. The footer of the login section reads 'OSIAS © 2018 Copyrights All Rights Reserved'.

Figure 5.11 Username invalid



The screenshot shows the same web application as Figure 5.11, but with a 'Password Invalid' error message. The username field now contains 'Gilbert136' and the password field is still masked with '****'. The 'Login' button remains at the bottom of the form. The sign-up section and footer are identical to Figure 5.11.

Figure 5.12 Password invalid

Register

Create your account

Username already exist

	Ben54
	Benjamin
	Franklin
	benFrank@gmail.com
	34565434
	Computer Sciences & Information Technology ▼
	Screenshot 2018-06-06 at 12.43.26 AM.png
	
	

Create Account

Log In

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Figure 5.13 Username already exit

Register

Create your account

Password do not match

 Ben54

 Benjamin

 Franklin

 benFrank@gmail.com

 34565434

 Computer Sciences & Information Technology

 Screenshot 2018-06-06 at 10.54.23 AM.png





Create Account

Log In

[OSIAS](#) © 2018 Copyrights All Rights Reserve

Figure 5.14 Password do not match

47

Register

Create your account

Add picture



hamsa34



Hamsah



Kame



hamsah@gmail.com



94038475



Communcation Studies



Please Click Here To Upload Your Profile Picture.....



....



....

Create Account

Log In

OSIAS © 2018 Copyrights All Rights Reserve

Figure 5.15 Add picture

Register

Create your account

Email is required

Email is not valid

First Name is required

Last Name is required

Contact is required

Username is required

Password is required

 Username

 Firstname

 Lastname

 Email

 Contact

 Communication Studies ▼

 Screenshot 2018-06-06 at 10.54.23 AM.png

 Password

 Repeat password

Create Account

Log In

OSIAS © 2018 Copyrights All Rights Reserve

Figure 5.16 Other security check

5.5 Deployment

This involves all the activities that make a software system available for use.

Software deployment may include activities such as release, installation and activation, deactivation, updates, version tracking and lastly adaptation.

CHAPTER SIX

SUMMARY, CONCLUSION AND RECOMMENDATION

6.1 Summary

Applications are critical to an organization's end-users' ability to perform their daily tasks. This section seeks to give a brief description of the capabilities, functionalities and usefulness of this software and the various steps and processes that are associated with it.

OSIAS is a chat application that uses web technologies which allow users to communicate and interact with each other without the use of the internet. It follows the client-server architecture, where there is a centralize system which is capable of handling client computers who want to connect to it in order to access certain resources or services.

The system only requires a minimum of two or more computers with at least a one gigabyte of memory and 856 megahertz of CPU clock speed each for the system to work effectively. Other secondary requirements may include a Wireless Access Point also known as Wi-Fi or an Ethernet system, and network configuration tools which are used by the administrator to maintain and control the system. A web browser and the known IP address of the system's server is also required for users

or clients to have access to the system. I recommend the use of Chrome or Mozilla web browser since that are known for their quality experience and performance.

What is the system capable of? The system is a chat application without the use of the internet. The system provides users or clients the ability to log in and sign up for an accounts which is used to identify them on the system. Also the system offers means of allowing users to interact and communicate with each other privately or in groups. Again, it allows users to send and receive media files.

What are some of the challenges? There are two main challenges of the system. The first is the size or capacity of the media files that can be sent or uploads. Currently, I have restricted the system to only allow up to 50 megabyte of data to be sent at a time. Lastly, the response time of the system, which now only takes a maximum of 10 seconds to update. This is all as a result of the computer that I am using, a Chromebook with a gigabyte of memory and 856 megahertz of CPU clock speed, it only offers the minimum requirement for the system to work and also because the system is under development which I soon intend to put into production, which will surely solve all the above challenges.

6.2 Conclusion

This system has come a long way, from its first inception to today where the system is now working and performing accurately and effectively. Currently, the system is

only capable of sending text and media files such as pictures, audios, videos and other document formats like PDF and etc. As of now, the system is still working perfectly and there has not being any critical known issues. It is in a good condition.

The first enhancement is to reduce the response time and also to increase the size of files that can be send across the system. I will also improve the user-interface design to make it easier for users to swiftly interact with the system.

Additional capabilities will be to add a blog section or component for users to post their own personal or necessary information, which can be a project they are working on, which will also be available to external stack-holders who might be interested in the activities of the employers and employees within a particular organization.

6.3 Recommendation

I will prefer this system to be used in any organization that is in need of a system that is capable of offering interactive and social services. The main aim of the system is to provide organization's members or employers and employees or other institutions the ability to interact and communicate with themselves at a lower cost as compared to that of using the internet. Also to allows for faster transmission of data and information between users of the system.

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APPENDIX A

USER'S MANUAL ORGANIZATION SOCIAL AND INTERACTIVE APPLICATION SYSTEM

Users of Organization Social and Interactive Application System are mostly employees and employers of any organization. This system can be used in academic institutions, corporate firms, religious institutions and many more.

1.0 GENERAL INFORMATION

This section explains the general terms of the system and the purpose for which it is intended.

1.1 SYSTEM OVERVIEW

Organization Social and Interactive Application System is an intranet chat application which uses web technologies. This system is developed to help employees and employers within an organization to interact and communicate with each other. This system also allows users of the system to share files or documents and other resources. This system also allows users to communicate privately with each other or communicate in groups.

1.2 ORGANIZATION OF THE MANUAL

The user's manual consists of five sections, these are the general information, system summary, getting started section, using the system and reporting.

The general information section explains in general terms the system and the purpose for which the system is intended.

The system summary section provides a general overview of the system. This outlines the uses of the system's hardware and software requirements, system's configuration, user access levels and system's behavior in case of any contingencies.

The getting started section explains how to get access to Organization Social and Interactive Application System and how to use and navigate around the system. This section presents a brief system menu. Also, this section provides a detailed description of system functions.

The reporting section describes the various ways information is collected by the application and the ways in which these information is presented and lastly, how to access these information.

2.0 SYSTEM SUMMARY

System summary section provides a general overview of the system. The summary outlines the uses of the system's hardware and software requirements, system's configuration, user access levels and system's behavior in case of any contingencies.

2.1 SYSTEM CONFIGURATION

Organization Social and Interactive Application System operates on windows OS, Android OS, Mac OS, Linux OS and many more. The application does not require access to the internet. File download and upload are all done in the intranet system. This system does not require any installation of any software, only a web browser is required, applicable Chrome browser. To access the system on the web browser, the IP address and port number of the computer hosting the system is required. This can be done by typing the IP address and the port number separated with a colon into the Universal Resources Locator in web browser.

2.2 USER ACCESS LEVELS

In Organization Social and Interactive Application System, anyone with valid credentials can use the application. This is done to help enforce security measure on the system.

2.3 CONTINGENCIES

In case of power outage, data are not saved in the internal memory of the operating device. This is one of the major situations that many cause the system to fail.

3.0 GETTING STATED

This section explains how to get access to Organisational Social and Interactive Application System. This section presents brief system menu.

3.1 INSTALLATION AND LOGGING IN

This system do not require any installation on the user end. The user only needs the IP address and port number of the computer hosting the system. This can be done by typing the IP address and the port number separated with a colon into the Universal Resources Locator in web browser. After that, the user will be redirected to the log in page where the user can provide the username or id and the password which will be authenticated, if there are no issues then the user will be granted access to use the system.

3.2 SYSTEM MENU

Organizational Social and Interactive Application System is in two sections. These section include the post page and the chat page. The post page allows users to post information in a group form. Every user is related to one of the three classes of groups, these include the public group, the department group and lastly, the private

group. Users can send text and media file in the post page. The chat page allows for private communication between individuals. Users can chat one-on-one with each other and can send or share media file.

3.3 EXIT SYSTEM

Users can exit from the system by logging out. Exiting the system, will take the user back to the login page and also delete the session information on the server side and the cookie and local storage information on the client side. This helps to safely clear all the user data from the system freeing memory space and tightening security.

4.0 REPORTING

The reporting section describes the various ways information is collected by the application and the ways in which these information is presented and lastly, how to access these information.

4.1 REPORTING CAPABILITES

Organizational Social and Interactive Application System provides reporting functionalities that allows users to know the status of a particular process. E.g. upon signing up for an account, a report message will alert the user of the state of the

process. A success report alerts the user that the particular actions is successful and a failure report, if the action is not successful.

APPENDIX B

SOURCE CODE SAMPLE

USER LOGIN COMPONENT

ORGANIZATIONAL SOCIAL AND INTERACTIVE APPLICATION SYSTEM

1.0 USER DATABASE MODEL

```
var mongoose = require('mongoose');

var bcrypt = require('bcryptjs');

// User Schema

var UserSchema = mongoose.Schema({

  username: {

    type: String,

    index:true

  },

  titles: {

    type: [String],

    required:false

  },

  firstname: {
```

```
    type: String,  
    required:true  
  },  
  lastname: {  
    type: String,  
    required:true  
  },  
  picture: {  
    type: String,  
    required:true  
  },  
  contact: {  
    type: String,  
    required:true  
  },  
  department: {  
    type: String,  
    required:true  
  },  
  password: {  
    type: String,  
    required:true
```

```
    },  
    email: {  
      type: String,  
      required: true  
    },  
    userstatus: {  
      type: String,  
      required: false  
    },  
    coursesduties: {  
      type: [String],  
      required: false  
    },  
    socialaccounts: {  
      type: [String],  
      required: false  
    },  
    profileinfo: {  
      type: [String],  
      required: false  
    },  
    date_created: {
```

```

    type: Date,

    default: Date.now

  }

});

var User = module.exports = mongoose.model('User', UserSchema);

//creating user

module.exports.createUser = function(newUser, callback){

  bcrypt.genSalt(10, function(err, salt) {

    bcrypt.hash(newUser.password, salt, function(err, hash) {

      newUser.password = hash;

      User.create(newUser, callback);

    });

  });

}

//getting user by name

module.exports.getUserByUsername = function(username, callback){

  var query = {username: username};

  User.findOne(query, callback);

}

```

```
//getting user detail by name //i dont know where this is being used
module.exports.getUserDetailByUsername = function(data, callback){
    var query = {username: data.username};
    var output = {password: 0};
    User.findOne(query, output, callback);
}
```

```
//update the user status if the user is online or offline
module.exports.updateUserStatus = function(data, userStatus, callback){
    var query = {_id: data._id};
    var value = {$set: {userstatus : userStatus}};
    User.update(query, value, callback);
}
```

```
//getting user by id used by deserialize //i dont know where this is being used
module.exports.getUserById = function(id, callback){
    User.findById(id, callback);
}
```

```
//compare password of user
module.exports.comparePassword = function(candidatePassword, hash,
callback){
    bcrypt.compare(candidatePassword, hash, function(err, isMatch) {
```

```

        if(err) throw err;

        callback(null, isMatch);

    });
}

//get all users

module.exports.getAllUsers = function(data, callback){

    //i have to get all users except the one who request for it, do this only if there is
    an ids query

    //i think i have solve it for now havent seen any errors yet

    var query;

    var output = {password: 0};

    //i have get all users if no data information is provided, which i have done but it
    is not very clear

    if(typeof data.ids === 'object' ){ query = { _id: { $nin: data.ids } } };

    }else if(typeof data.ids === 'string'){ query = { _id: { $ne: data.ids } } };

    User.find(query, output, callback);

}

//compare if username already exist

module.exports.checkUser = function(data, callback){

    var query = {username: data};

```

```
var output = {password: 0};

User.findOne(query, output, callback);

}
```

2.0 USER SERVER CODES

```
//login function

//i have go and read into this further

module.exports.Login = function(req, res, next) {

  passport.authenticate('local',

    function(err, user, info){

      if(err){ return next(err);}

      if(!user){ return res.send(info);}

      req.logIn(user, { "session" : false }, function(err){

        if(err){ return next(err);}

        return res.send(

          {success: {

            id: user.id,

            username: user.username,

            email: user.email,

            picture: user.picture,

            firstname: user.firstname,

            lastname: user.lastname,
```



```

        contact: user.contact,

        department: user.department}

    });

    });

    })(req, res, next);

    //will create a hash function

};

```

3.0 USER CLIENT LOGIC CODES

```

import { Component } from '@angular/core';
import { Router } from '@angular/router';
import { UsersService } from '../users.service';
import { sharedFunctions } from '../shared/sharedFunctions';

```

```

@Component({

    templateUrl: 'login.component.html'

})

export class LoginComponent {

    username: string;

    password: string;

    userInfo: any[];

```

```
error: string;
```

```
constructor(private userservice : UsersService, private router : Router, public  
sharedFunctions: sharedFunctions) { };
```

```
authuser(){
```

```
    const checkUser = {  
        username : this.username,  
        password : this.password  
    }  
}
```

```
    this.userservice.checkUser(checkUser)
```

```
        .subscribe(info => {
```

```
            this.userInfo = info['success'];
```

```
            if(info['success']){
```

```
                this.sharedFunctions.showImage(this.userInfo, 'users', 'image',  
'profilePictures');
```

```
                this.storeLocalStorage();
```

```
            }else if(info['message']){
```

```
                this.error = info['message'];
```

```
            }
```

```
        })
```

```
};
```

```

storeLocalStorage(){
  try{
    if(typeof(this.userInfo['image']) == "undefined"){
      throw new Error('Please wait');
    }else{
      //i have to encrypt userdetails and store in the localStorage
      localStorage.setItem( 'userDetail', JSON.stringify(this.userInfo));
      this.router.navigate(['/mypage', this.userInfo['username']]);
    }catch(e){
      setTimeout(()=>{ this.storeLocalStorage()}, 5000);}
  }
}

```

4.0 USER CLIENT PRESENTATION CODES

```

<div class="app flex-row align-items-center">
  <div class="container">
    <div class="row justify-content-center">
      <div class="col-md-8">
        <div class="card-group mb-0">
          <div class="card p-4">
            <div class="card-body">
              <form (submit) = "authuser()">

```

```
<h1>Login</h1>
```

```
<p class="text-muted">Sign In to your account</p>
```

```
<div *ngIf = "error" >
```

```
  <div class = "alert alert-danger">
```

```
    {{error}}
```

```
  </div>
```

```
</div>
```

```
<div class="input-group mb-3">
```

```
  <span class="input-group-addon"><i class="icon-  
user"></i></span>
```

```
    <input type="text" class="form-control" [(ngModel)] =  
"username" name = "username" placeholder="Username">
```

```
</div>
```

```
<div class="input-group mb-4">
```

```
  <span class="input-group-addon"><i class="icon-  
lock"></i></span>
```

```
    <input type="password" class="form-control" [(ngModel)] =  
"password" name = "password" placeholder="Password">
```

```
</div>
```

```
<div class="row">
```

```
  <div class="col-6">
```

```

        <input type="submit" class="btn btn-primary px-4"
value="Login">

    </div>

    <div class="col-6">

        <button type="button" class="btn btn-primary d-lg-none
float-right" [routerLink]="['/pages/register']">Register Now!</button>

    </div>

</div>

</div>

</form>

<div class="text-center mt-4">

    <a href="#">OSIAS&nbsp;</a>

    <span class="text-muted">&copy; 2018 &nbsp;& Copyrights All
Rights Reserved</span>

</div>

</div>

</div>

<div class="card text-white bg-primary py-5 d-md-down-none"
style="width:44%">

    <div class="card-body text-center">

        <div>

            <h2>Sign up</h2>

```

Welcome to OSIAS, Sign Up to be a member of this organisation. You can only register once. Please obey the law. If there is any problem, see one of the administrators. Thank you

[Register Now!](#)

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